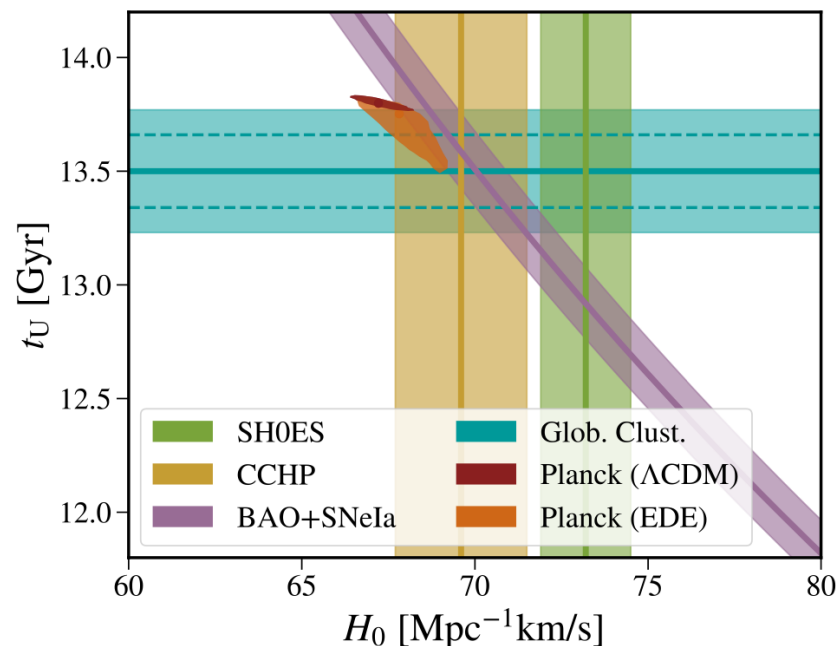




Raul Jimenez



CosmoAges: what will be the next disruption?



ERC Synergy RedH0t



European Research Council
Established by the European Commission

<https://www.ias.edu/simonsgeometry>



Λ CDM: The standard cosmological model

Just 7 numbers.....

describe the Universe composition and evolution

Homogenous background

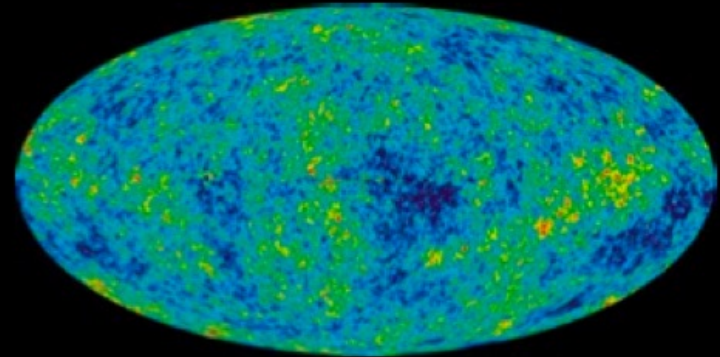
Perturbations



$\Omega_b, \Omega_c, \Omega_\Lambda, H_0, \tau$

- atoms 4%
- cold dark matter 23%
- dark energy 73%

$\Lambda?$ CDM?



A_s, n_s, r

- nearly scale-invariant
- adiabatic
- Gaussian

ORIGIN??

Move **BEYOND** parameter **FITTING: TEST the MODEL**

What We Should Be Doing is Selecting The Right Model

Abstract

1 Introduction

2 The Coherence Principle: A general method for physical priors from structural grammar

3 The Coherence Principle in practice

References

The Coherence Principle: A Falsifiable Prior for Model Selection from the Grammar of Theories

Raul Jimenez

Carlos Peña Garay

Fergus Simpson

Licia Verde

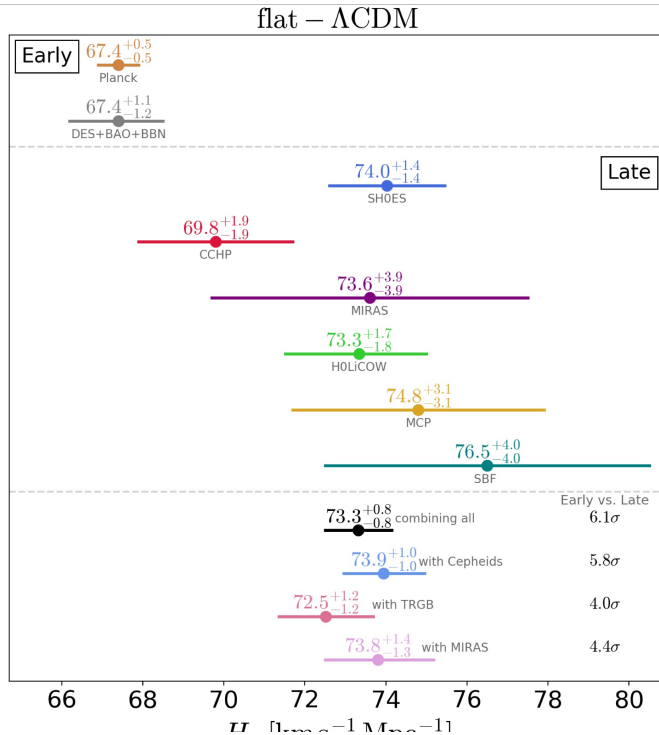
Abstract

Bayesian model selection in cosmology and particle physics is often performed in regimes where the posterior odds inherit a strong and frequently unacknowledged dependence on the prior assigned to the space of competing models. Standard responses, such as reference priors, hierarchical priors, or appeals to naturalness, either ignore relevant theoretical knowledge or rely on criteria that are hard to define operationally.

We propose the *Coherence Principle*: a reproducible way to assign model priors according to their compatibility with the validated structure of an existing theory. This structure, or *grammar*, includes symmetries, conservation laws, locality, Lorentz invariance, and universality patterns. Unmotivated violations of these rules carry a coherence cost. This cost is converted into a prior weight through a maximum-entropy exponential form, controlled by a single calibratable parameter α . The resulting prior is distinct from both the Bayesian Occam factor and naturalness.

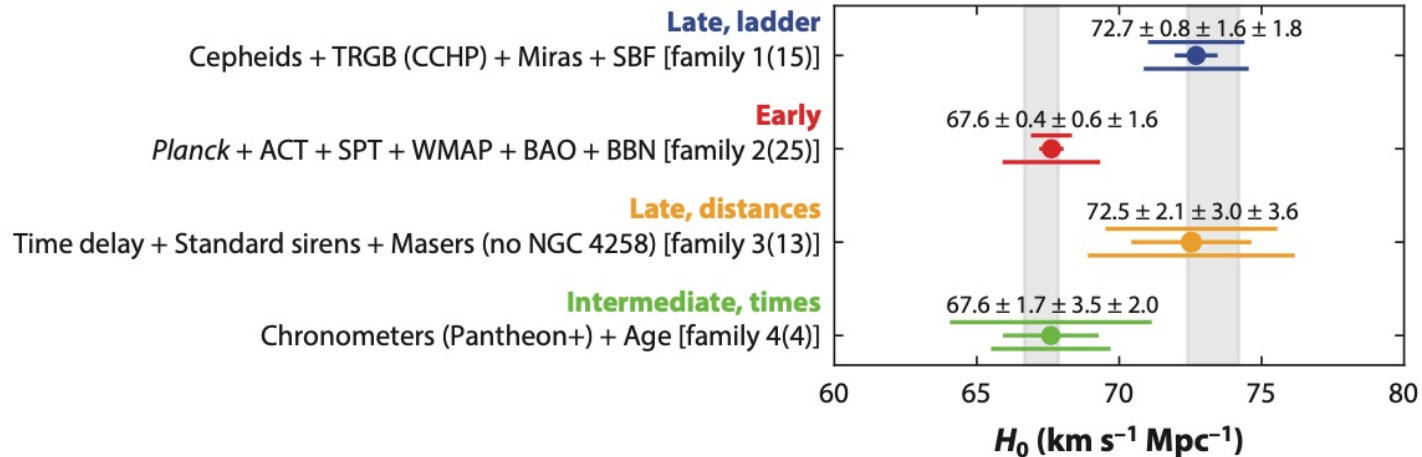
We illustrate the application of the principle using examples drawn mostly from cosmology, namely neutrino mass mechanisms, dark energy and modified gravity, inflation, beyond-Standard-Model sectors, and hierarchical astrophysical inference. We also test it on historical cases: general relativity, Pauli's neutrino, parity violation, and special relativity. These examples show that the principle favors the historically successful choice when the relevant theoretical grammar is defined in the correct domain and at the correct time.

Tensions within the model



From Verde et al. Nature Astronomy (2019)

Same situation 6 years later
(from Verde et al. ARAA 2025)



H_0 AND ODDS ON COSMOLOGY

ANDREW JAFFE

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Received 1995 September 18; accepted 1996 April 16

ABSTRACT

Recent observations by the *Hubble Space Telescope* of Cepheids in the Virgo cluster imply a Hubble constant $H_0 = 80 \pm 17 \text{ km s}^{-1} \text{ Mpc}^{-1}$; other recent observations find $70 \lesssim H_0 \lesssim 90 \text{ km s}^{-1} \text{ Mpc}^{-1}$, with several large excursions in either direction. We attempt to clarify some issues of interpretation of these results for determining the global cosmological parameters Ω and Λ . In this paper, we use these results as a case study in the formalism of Bayesian model comparison, allowing a rigorous comparison of the different cosmological possibilities. We concentrate our analysis on three recent determinations of the Hubble constant, but the results are generic so long as they prefer $H_0 t_0 \gtrsim 1$, which would seem to require $\Lambda > 0$ within the context of Friedmann-Robertson-Walker cosmologies. With our more rigorous methods, the data do indeed suggest a universe with a nonzero cosmological constant but vanishing curvature: $\Omega + \Lambda = 1$.

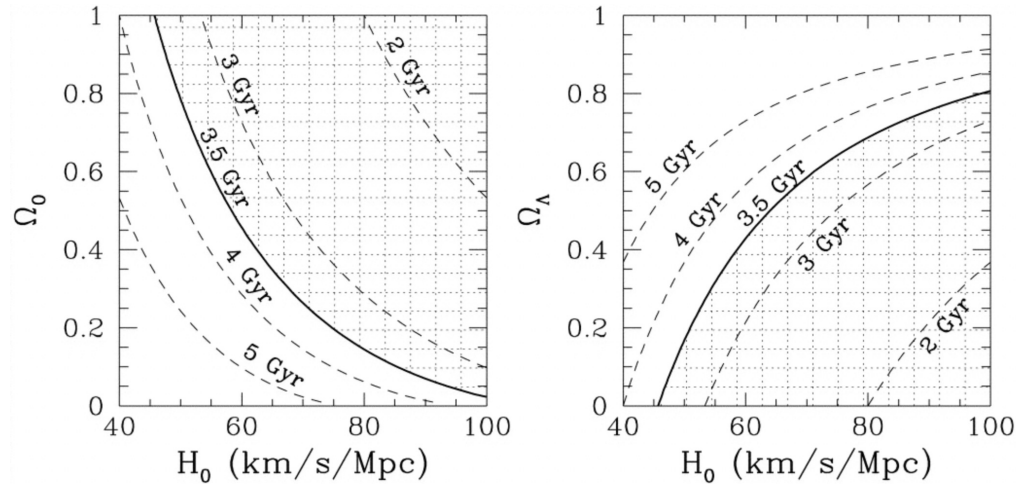
marginalize over at least a subset of these. To use Bayes's theorem, we will need to assign a prior probability to these parameters, $p(H_0 t_0 \tau \Omega \Lambda | I)$. Obviously, there are several relationships among the parameters that we must consider. First, if t_0 is the age of the universe, then we know $t_0 > \tau$. Second, the age and Hubble constant are related to the cosmological parameters in a matter-dominated FRW universe by

$$\begin{aligned} H_0 t_0 &= 1.02h \frac{t_0}{10 \text{ Gyr}} = f(\Omega, \Lambda), \\ &= \int_0^1 dx \frac{1}{\sqrt{1 - \Omega - \Lambda + \Omega x^{-1} + \Lambda x^2}}, \end{aligned} \quad (5)$$

(Riess et al. 1994). In addition, we care about the measured age of some “old” object in the universe, $\hat{\tau} \pm \sigma_\tau$. Recent determinations prefer a globular cluster age of $\tau_{gc} = 14.69 \pm 2 \text{ Gyr}$ (Chaboyer 1995), which we use in the following.

In fact absolute stellar ages at $z=0$ and $z=1.5$ were **alone giving evidence of dark energy in 1995**

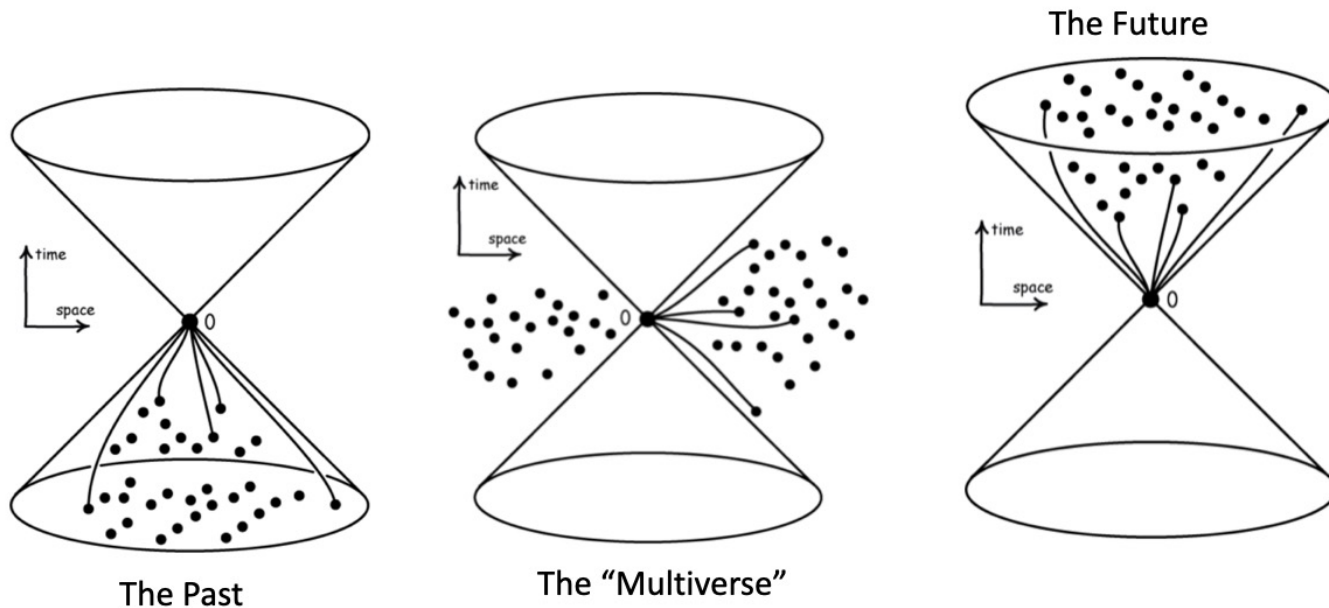
Age of radio galaxy LBDS53W091 is > 3.5 Gyr (Dunlop et al. 1996)



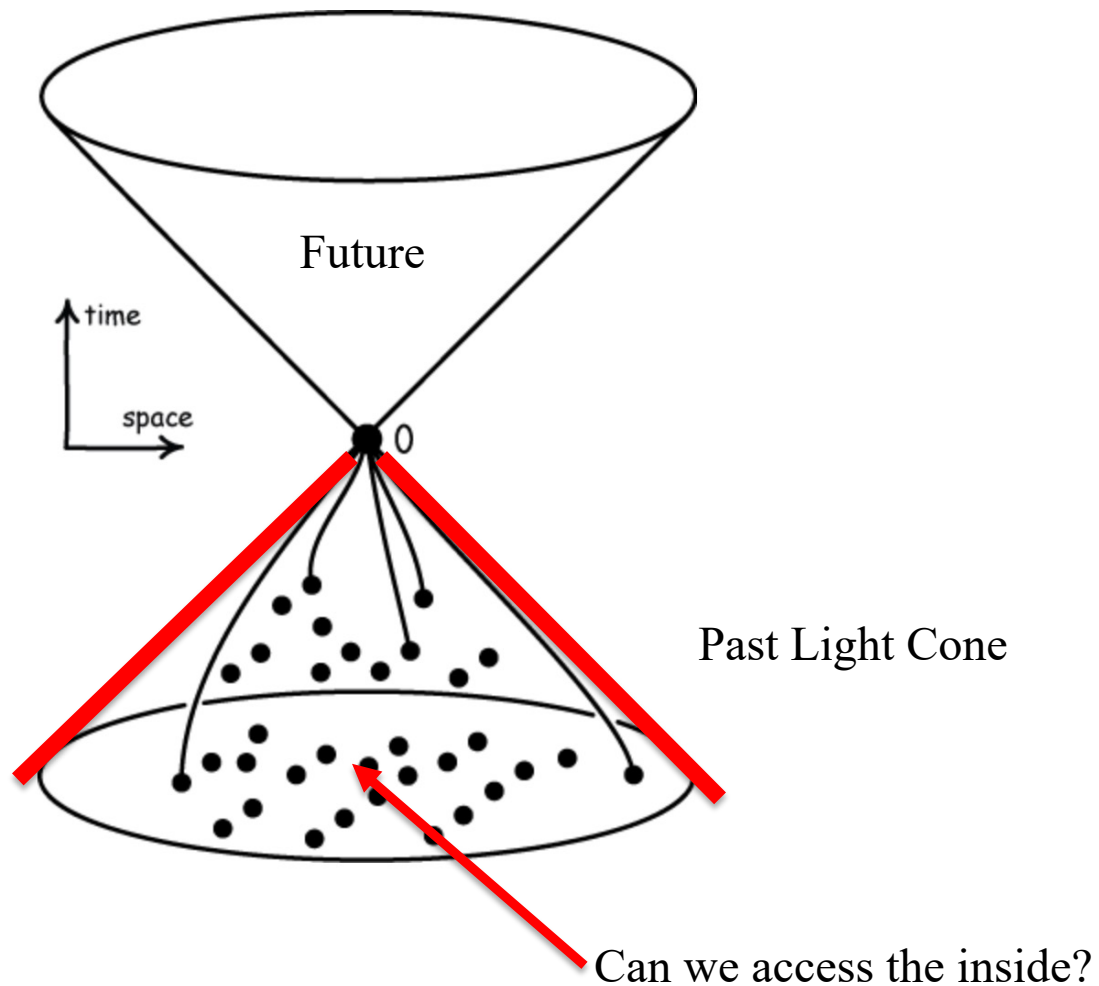
Ages of GCs > 12 Gyr @ 95% CL (Jimenez et al. 1993) only $H_0 > 60$ fits... $\Lambda > 0$!

From Jimenez, Uffe Jorgensen et al. 1995

An exploration of the light cone: how do we do it?



All that BAO, CMB, SN see/measure is the surface of the past light cone



Stellar ages: a tool to measure the expansion rate and test the cosmological model!

- Relative stellar ages at z provide an estimate of the expansion rate $H(z)$

$$\delta t(z) \simeq \frac{\delta z}{H(z)(1+z)}$$



Cosmology independent

Absolute stellar ages

- Absolute stellar ages (clocks) at $z=0, 5, 10\dots$ provide a cross-check of the model

$$H_0 = \frac{A}{t} \int_0^{z_t} \frac{1}{1+z} \left[\Omega_{m,0}(1+z)^3 + (1 - \Omega_{m,0})(1+z)^{3(1+w)} \right]^{-1/2} dz$$



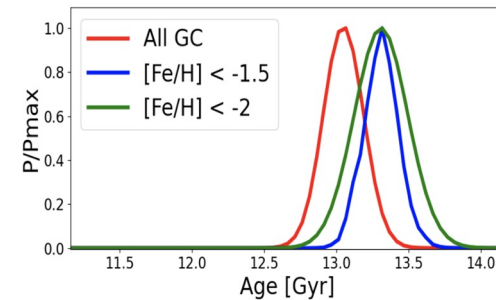
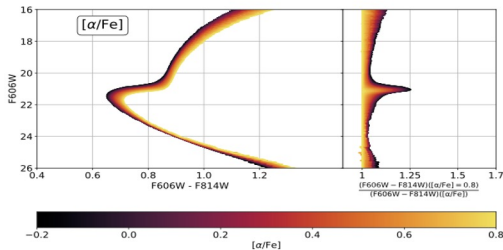
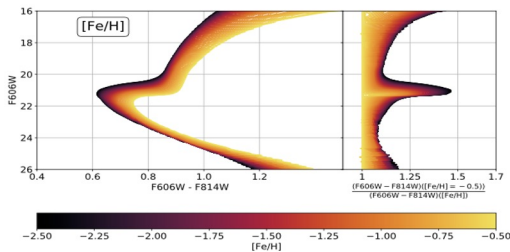
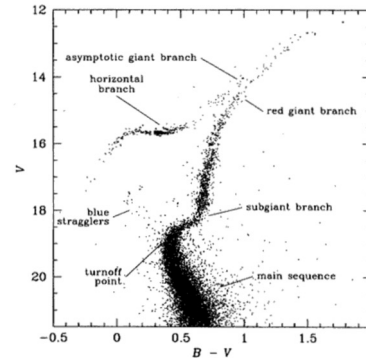
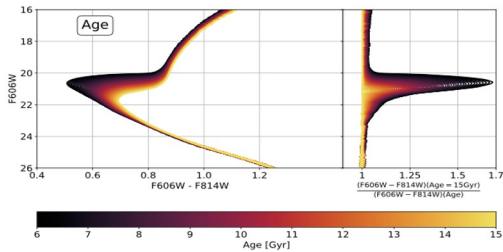
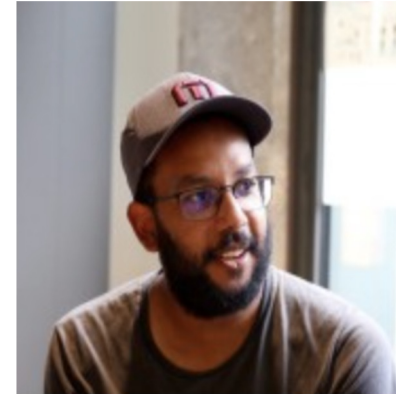
Relies on knowing other background cosmological parameters
(or the expansion history “shape”)

“The local and distant Universe, stellar ages and H_0 ”
JCAP 2019 ,Jimenez, Cimatti, Verde, Moresco, Wandelt

“Revisiting the Oldest Stars as Cosmological Probes: New Constraints on the Hubble Constant” ApJ 2023, Cimatti & Moresco

A full Bayesian estimate of GCs parameters

Valcin et al. JCAP 2021; 2025

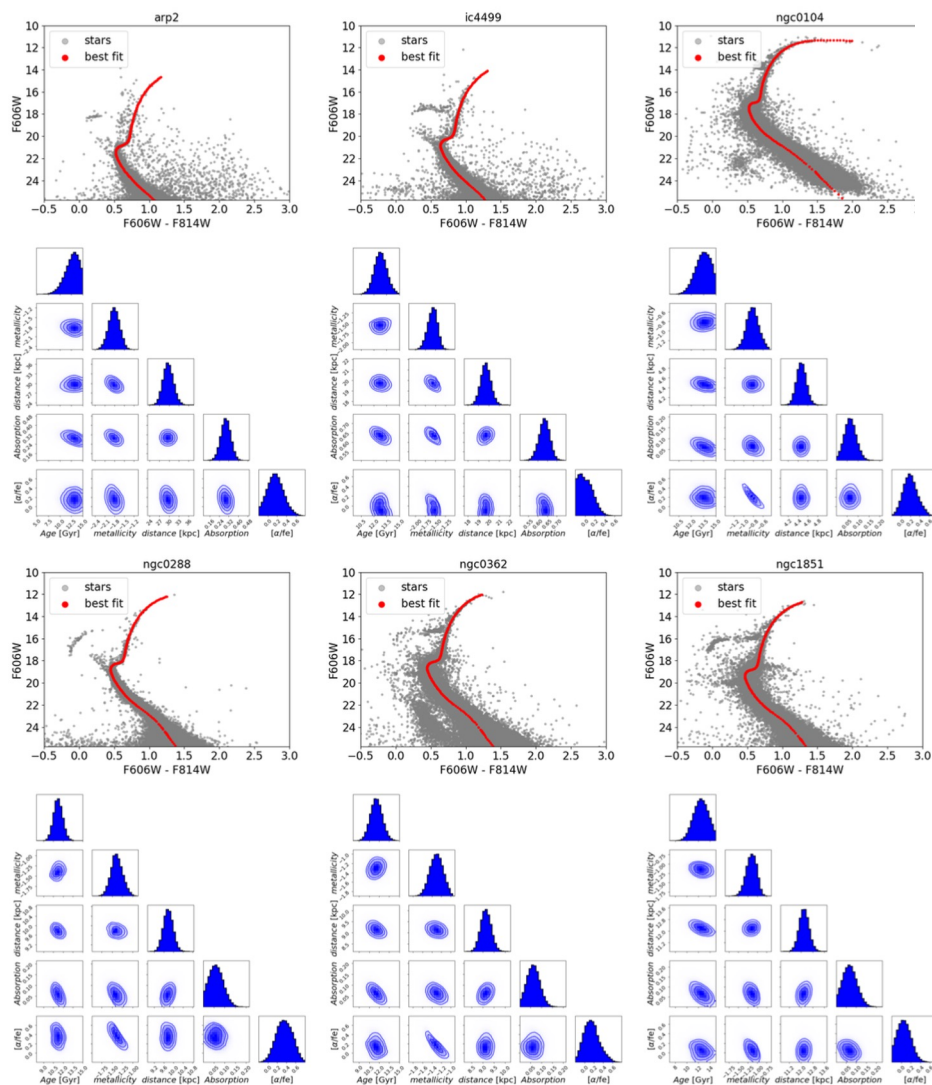


$$t_U = 13.57^{+0.16}_{-0.14}(\text{stat}) \pm 0.23(\text{sys.})$$

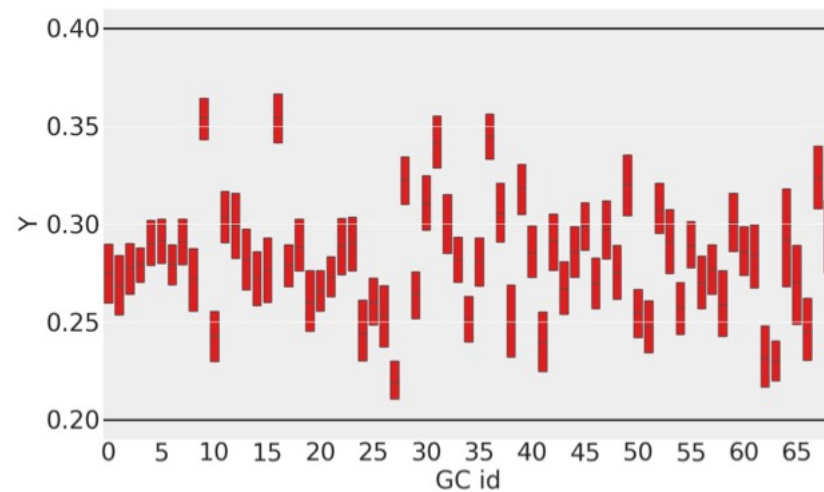
$$t_U^{\leq} = 13.92^{+0.13}_{-0.1}(\text{stat}) \pm 0.23(\text{sys.})$$

Free parameters: age, distance, [Fe/H], [alpha/Fe], Helium, Reddening, dust and Y1 and Y2

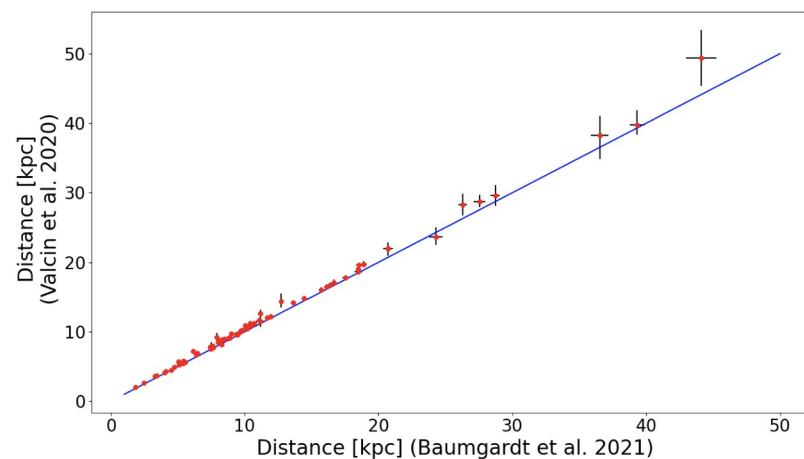
Contours DO close



Recovered Helium content



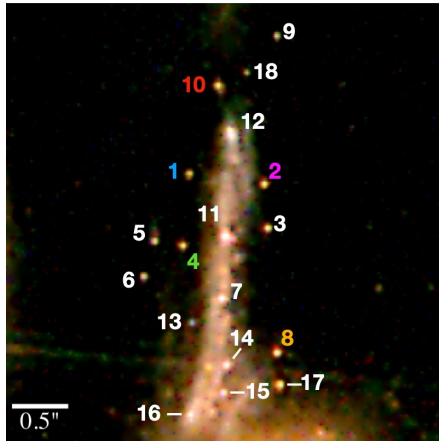
Distances DO agree with GAIA's



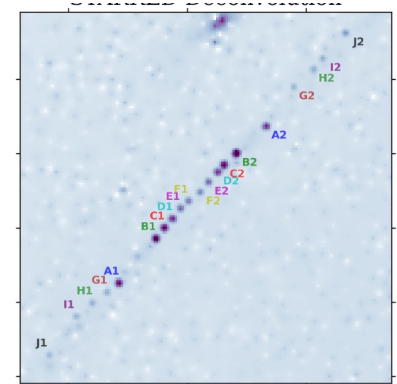
Ages of GCs for $z \gg 0$

Tomasetti et. al. A&A 2024 & 2025 & 2026

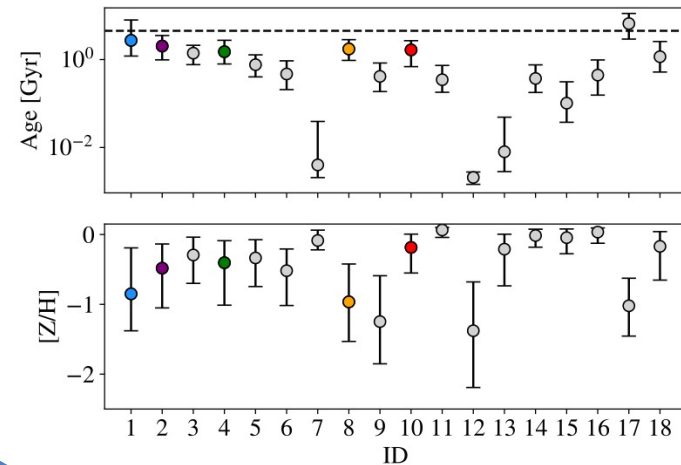
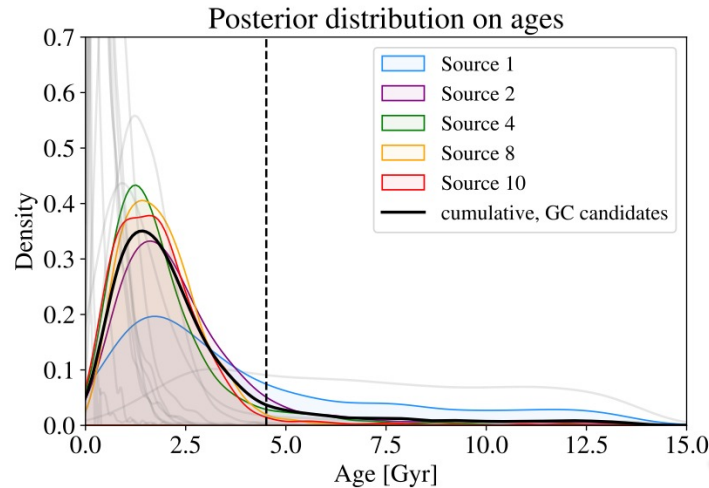
Sparkler $z = 1.4$



GEMS $z=10$



Thanks to new image deconvolution pipeline
the analysis **does not** use any cosmology prior



Prior on age

Cosmic Chronometers

differential ages

Simon et al. 2005, Stern et al. 2010, Moresco et al. 2012, 2015, 2016



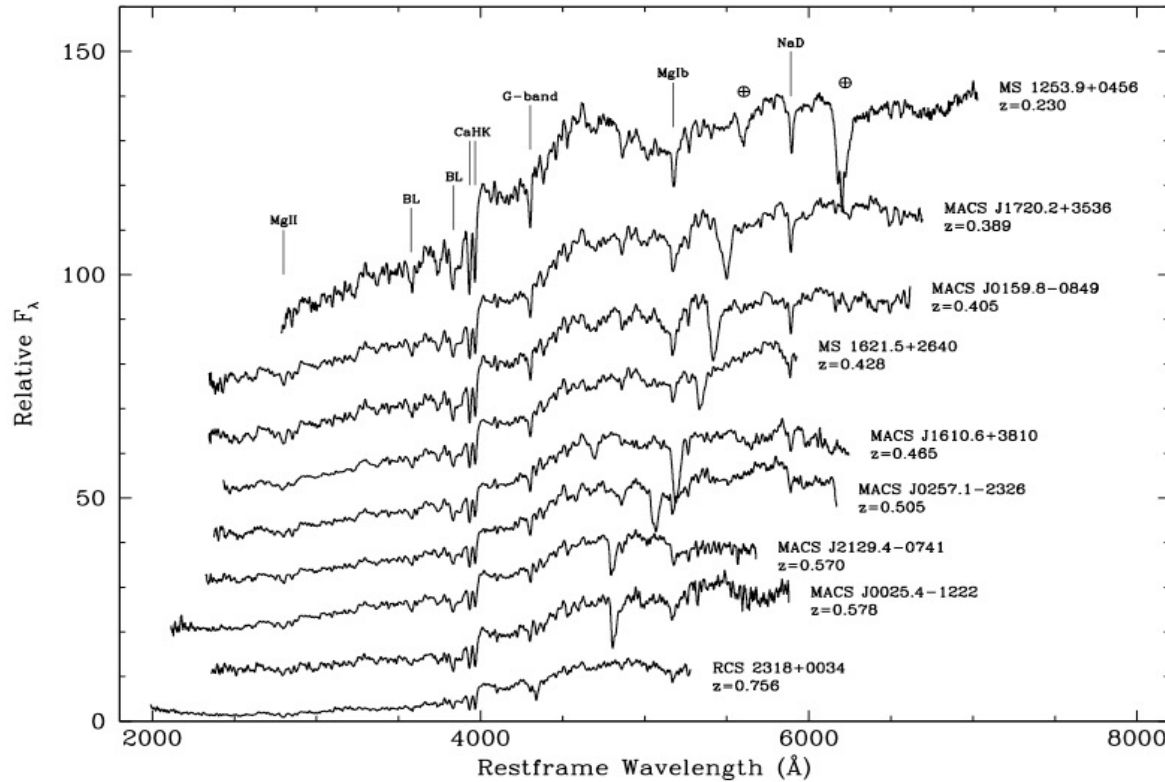
Old elliptical galaxies have stellar populations well described by a single age.

Differential ages give

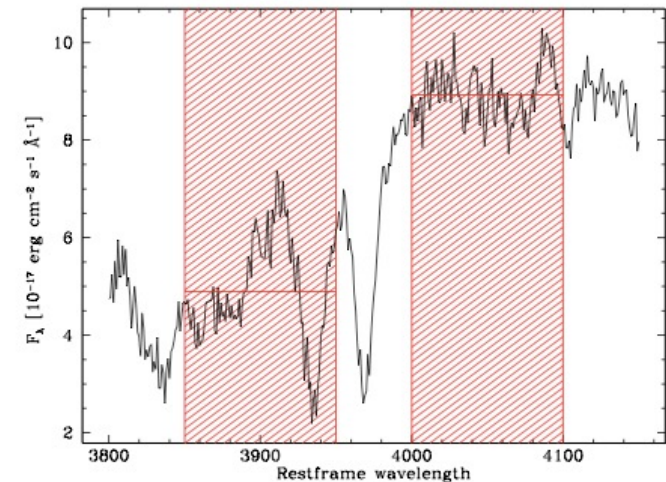
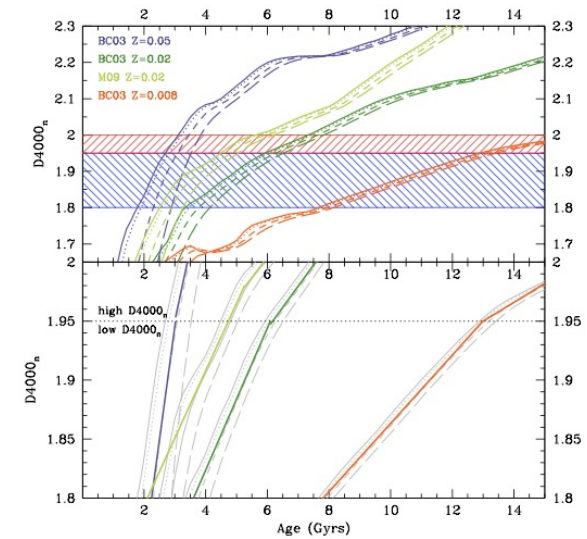
$$\delta t(z) \simeq \frac{\delta z}{H(z)(1+z)}$$

(ignores spread in formation time in comparison with Hubble time)

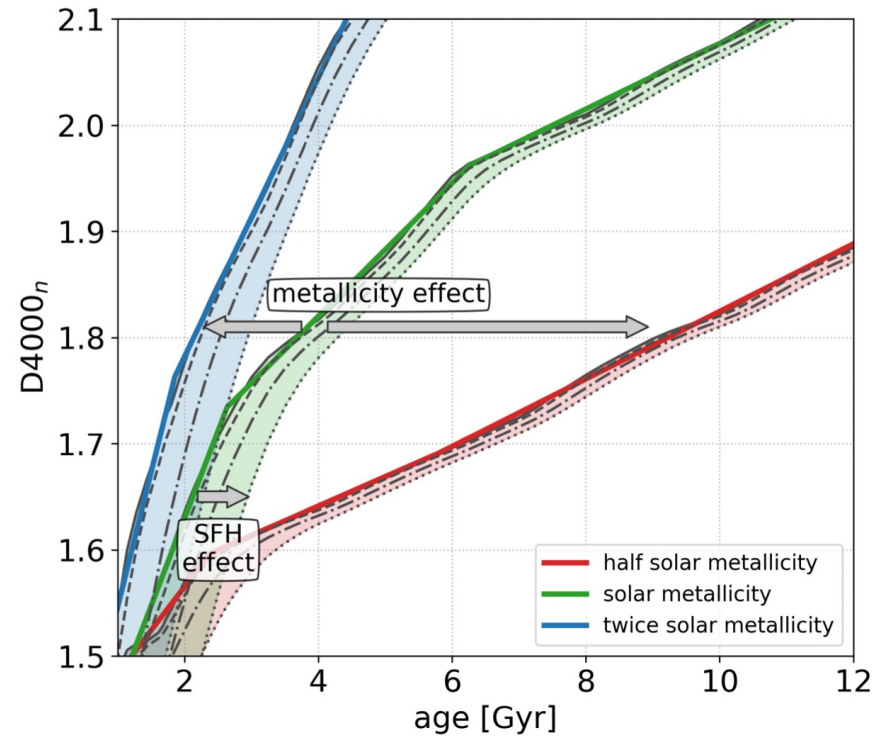
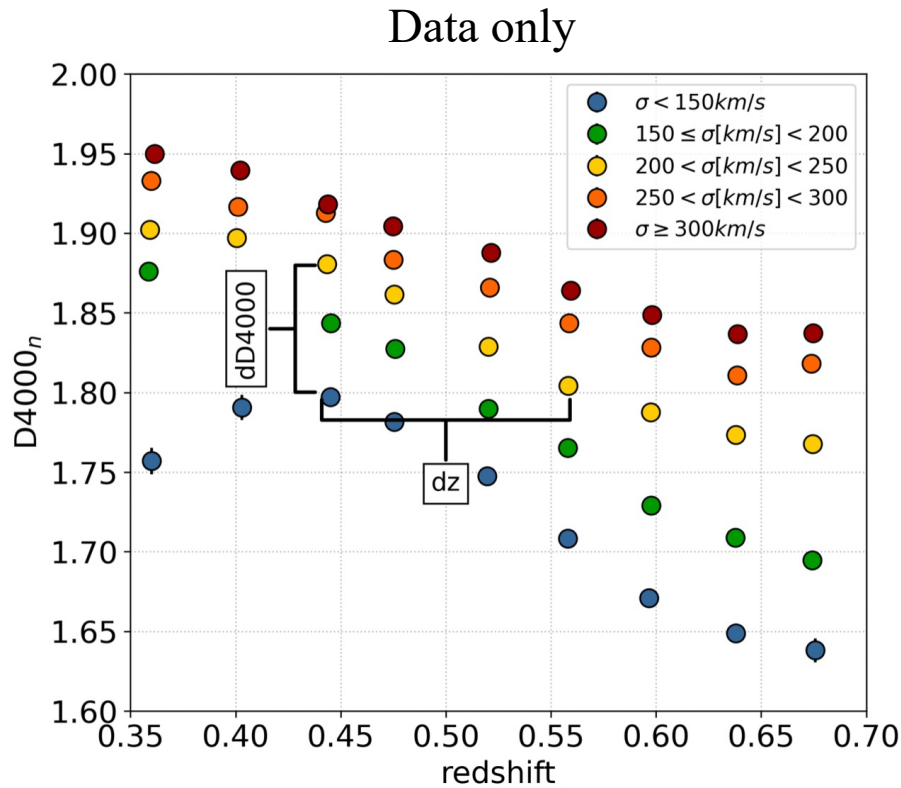
Relative aging of galaxies using D4000 break



$$H(z) = -\frac{A}{1+z} \frac{dz}{dD4000_n}$$



D4000 as chronometer



See review by Moresco et al. arXiv 2201.07241 in press Living Reviews in Relativity

Relative Aging of Galaxies

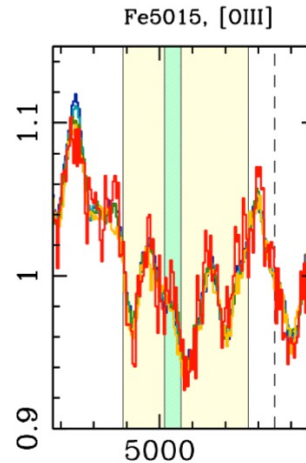
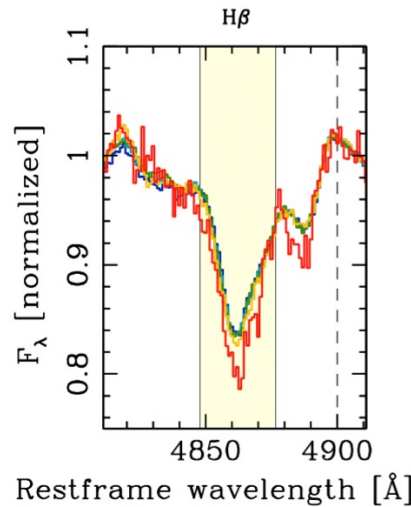
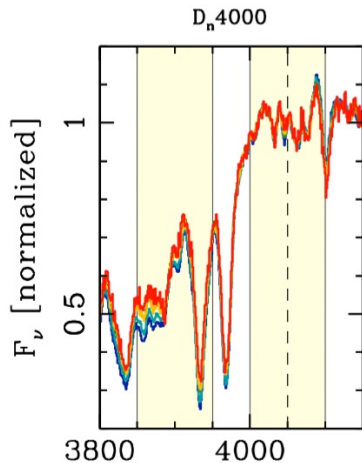
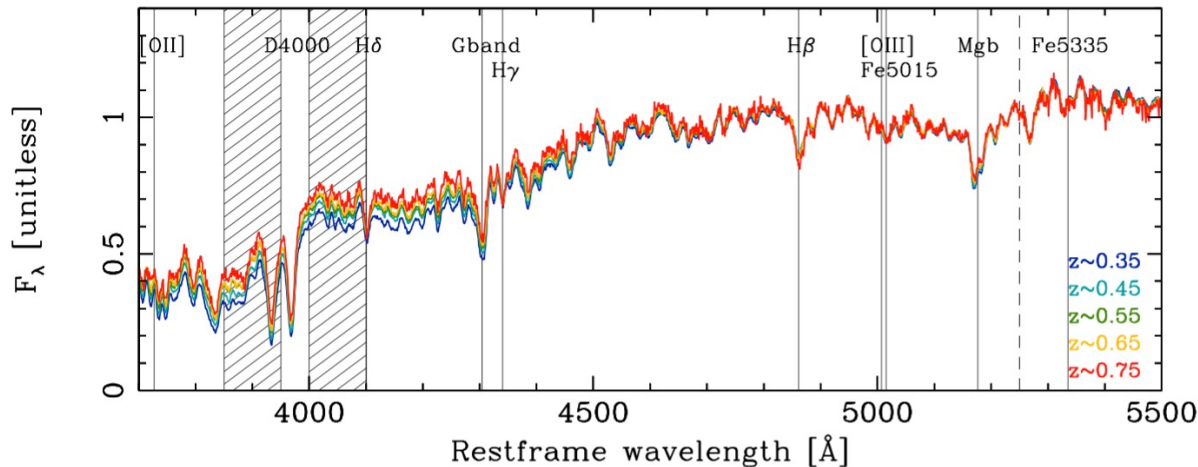


TABLE 6
METALLICITY SENSITIVITIES

Index	$\left(\frac{\Delta \text{age/age}}{\Delta Z/Z} \right)_I$
$U - V$	1.5
$B - V$	1.4
$V - R_c$	1.3
$V - I_c$	1.4
$V - J$	1.9
$V - K$	1.9
$J - H$	1.7
$J - K$	1.9
$J - L$	1.8
$J - L'$	1.8
$J - M$	1.7
01 CN ₁	1.9
02 CN ₂	2.1
03 Ca4227	1.5
04 G4300	1.0
05 Fe4383	1.9
06 Ca4455	2.0
07 Fe4531	1.9
08 Fe4668	4.0
09 H β	0.6
10 Fe5015	4.0
11 Mg ₁	1.8
12 Mg ₂	1.8
13 Mg <i>b</i>	1.7
14 Fe5270	2.3
15 Fe5335	2.8
16 Fe5406	2.5
17 Fe5709	6.5
18 Fe5782	5.1
19 Na D	2.1
20 TiO ₁	1.5
21 TiO ₂	2.5
D(4000)	1.3

Worthey 1994; Lick indices

Hbeta is the least sensitive Lick-index to metallicity: just tracks age. While Fe5015 tracks metallicity. High-*z* galaxies are older, while all galaxies have same metallicity.

Thou Shall Use The Full Covariance Matrix

Available Moresco et al. (2020)

<https://gitlab.com/mmoresco/CCcovariance>

[Why GitLab](#) [Pricing](#) [Explore](#) [Sign in](#) [Get free trial](#)

Project

C Cosmic chronometers covariance estimate

Manage >

Plan >

Code >

Deploy >

Operate >

Monitor >

Analyze >

Michele Moresco / Cosmic chronometers covariance estimate

C Cosmic chronometers covariance estimate

☆ Star 4

master CCcovariance

Find file

Code

mc Michele Moresco authored Mar 19, 2021

88141333

History

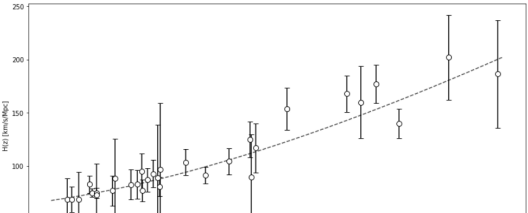
Name	Last commit	Last update
data	Initial commit	4 years ago
examples	Initial commit	4 years ago
figures	Initial commit	4 years ago
README.md	mc	4 years ago

README.md

Cosmic chronometers Covariance

A primer on how to estimate cosmic chronometers covariance, how to decouple it into its various components, and how to apply it for cosmological calculations

- Dependencies:
 - Python 3
 - numpy
 - scipy



Project information

A primer on how to estimate cosmic chronometers covariance, how to decompose it in the various components, and how to use it for cosmological calculation

2 Commits

1 Branch

0 Tags

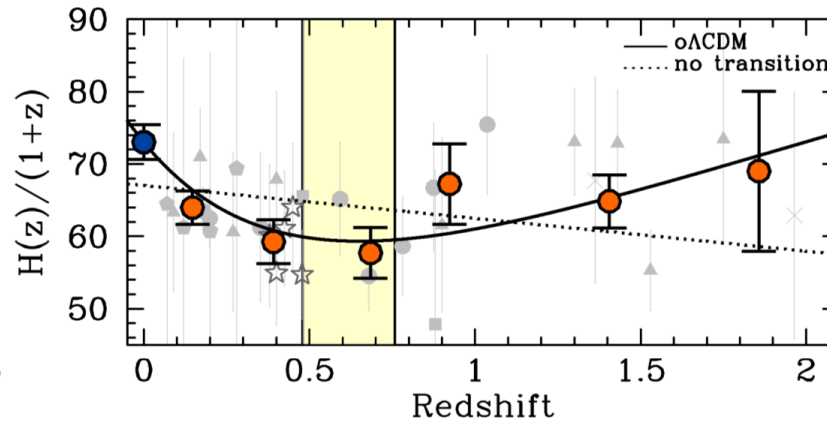
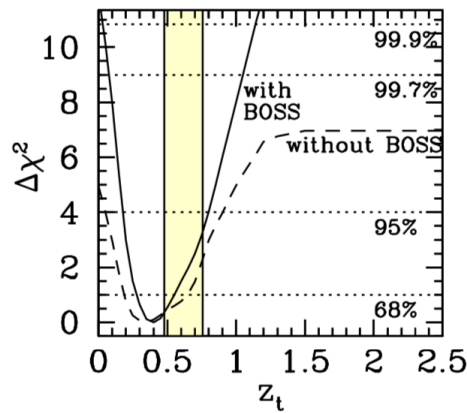
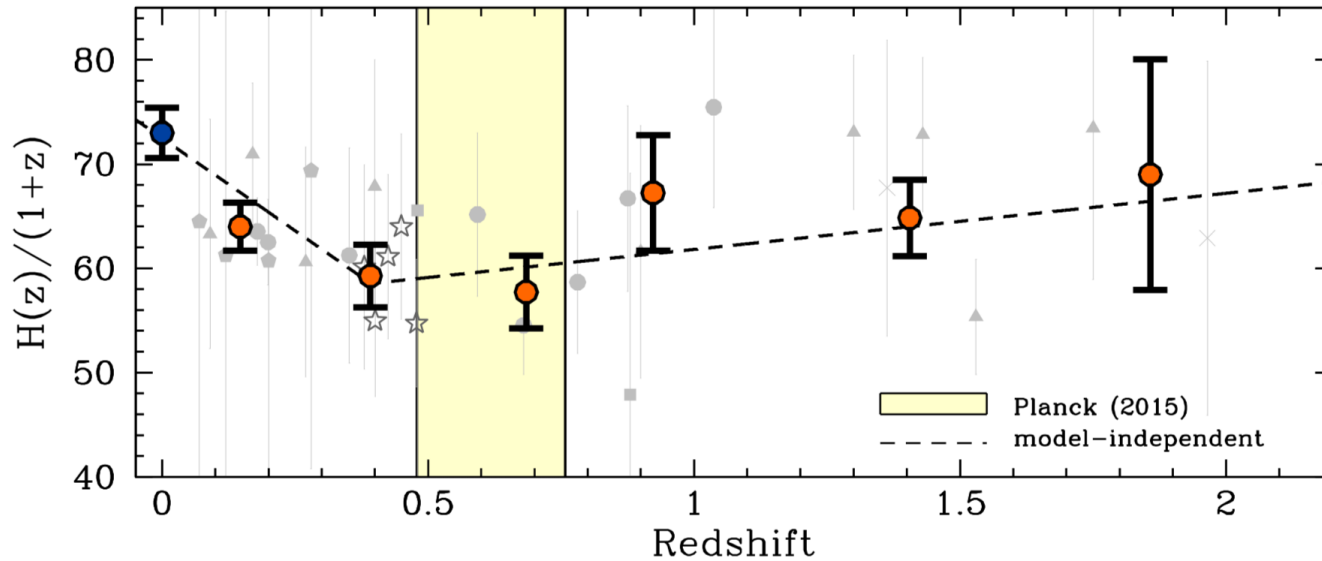
README

Created on

March 18, 2021

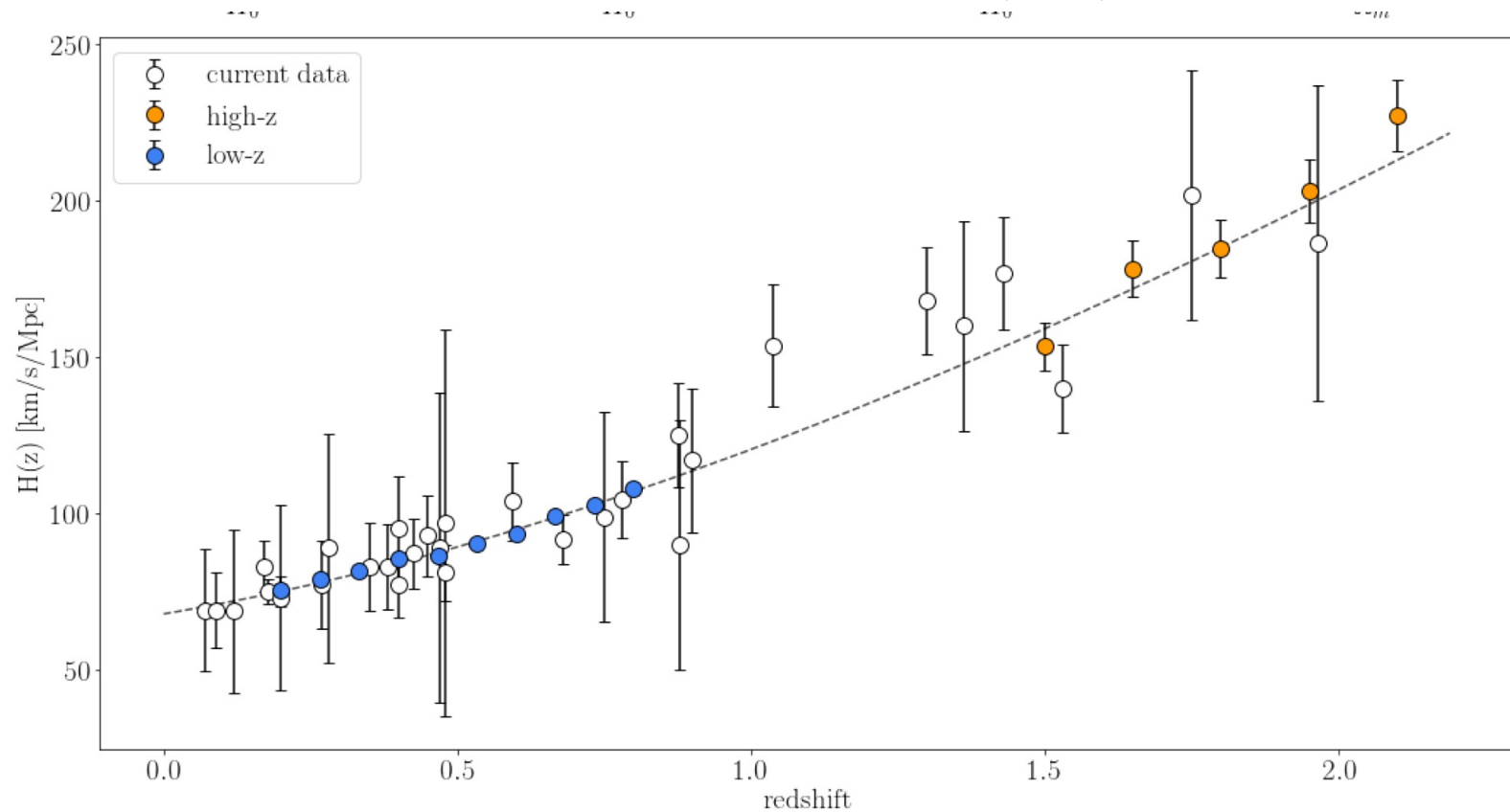
[Help](#)

Some results (No Cosmology Assumed)



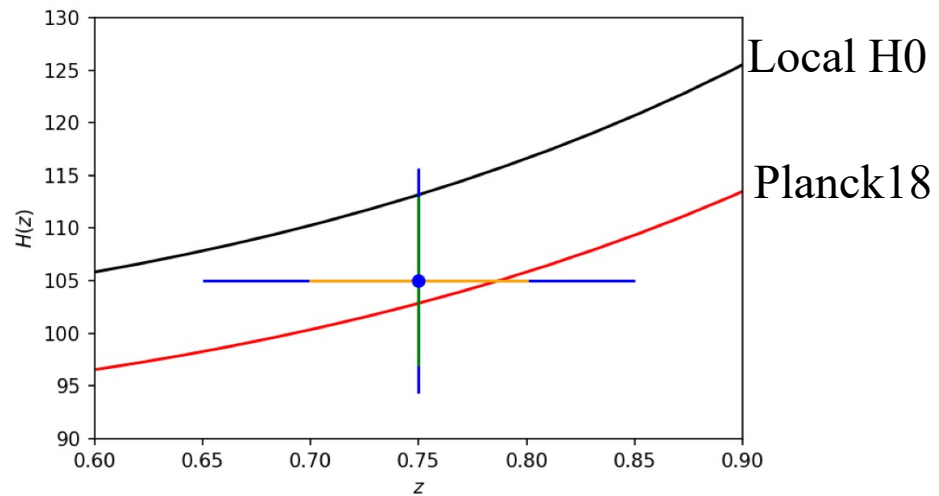
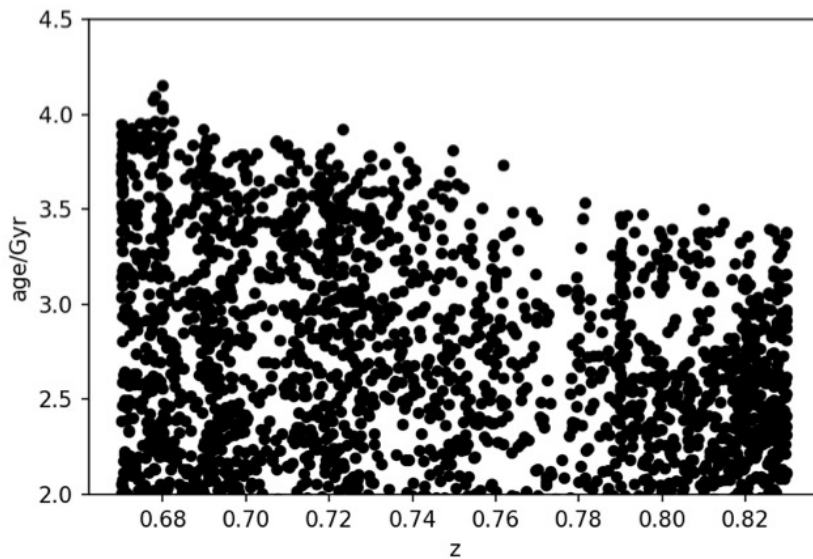
How much better can you do with EUCLID/DESI?

Forecast Moresco et al. (2023)



Cosmic Chronometers with Photometry only

Available data

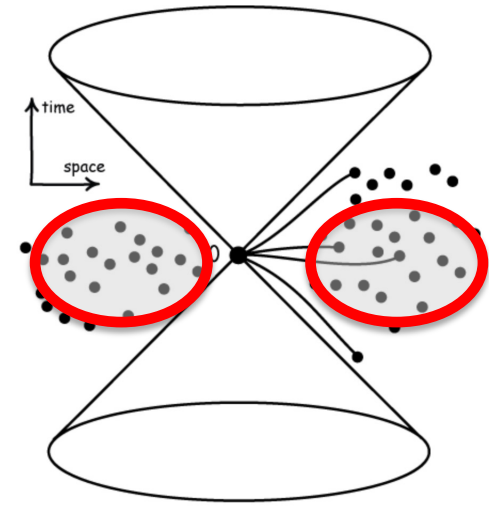
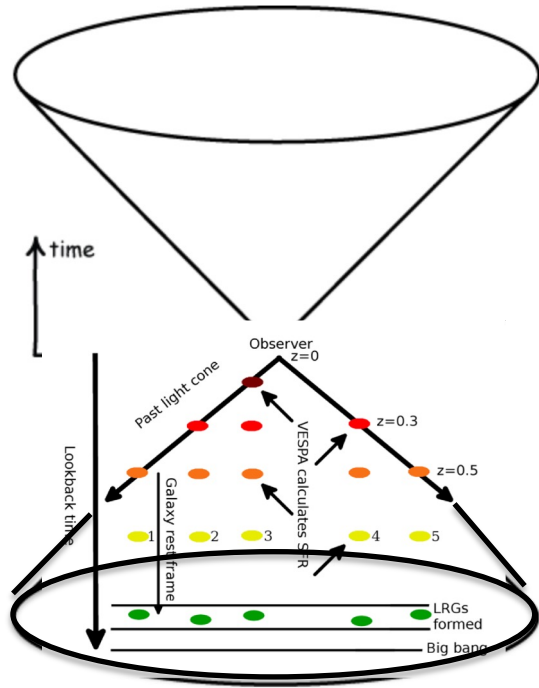


Jimenez et al. JCAP 2023

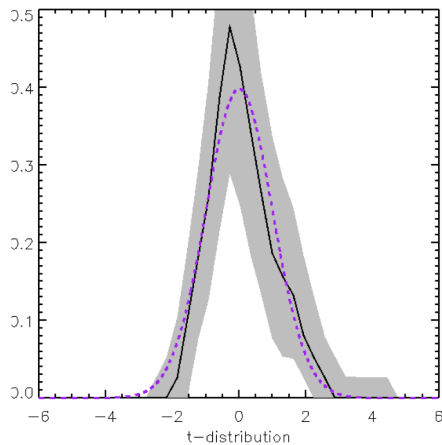
Opens prospects for massive LSST/EUCLID cosmic-chronometers

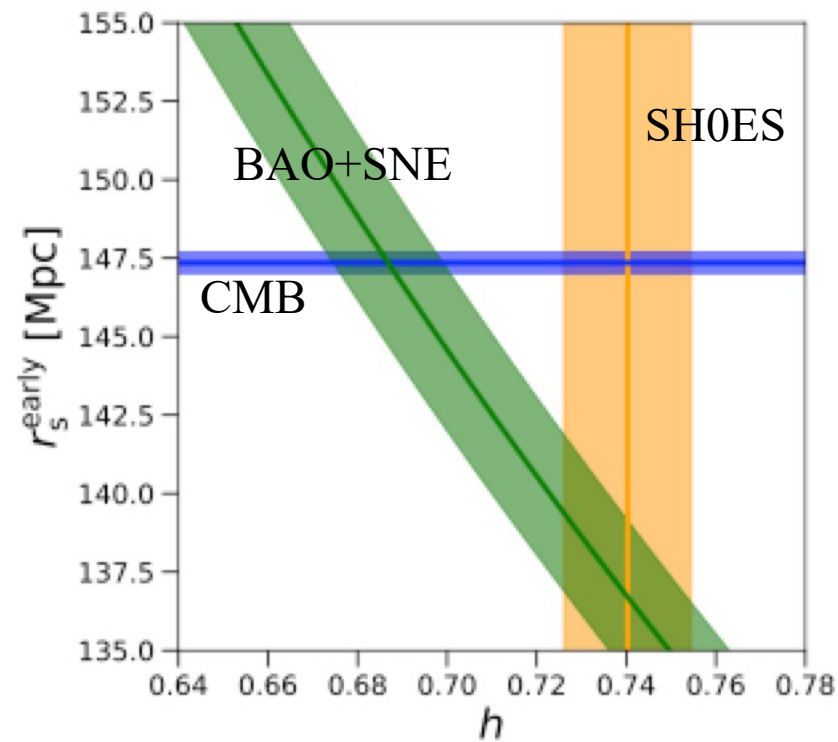
Measuring Homogeneity

(Hoyle et al. JCAP 2012) [Current Data]



Peering Beyond the Horizon (Jimenez et al. JCAP 2012; JCAP 2022) [Forecast]





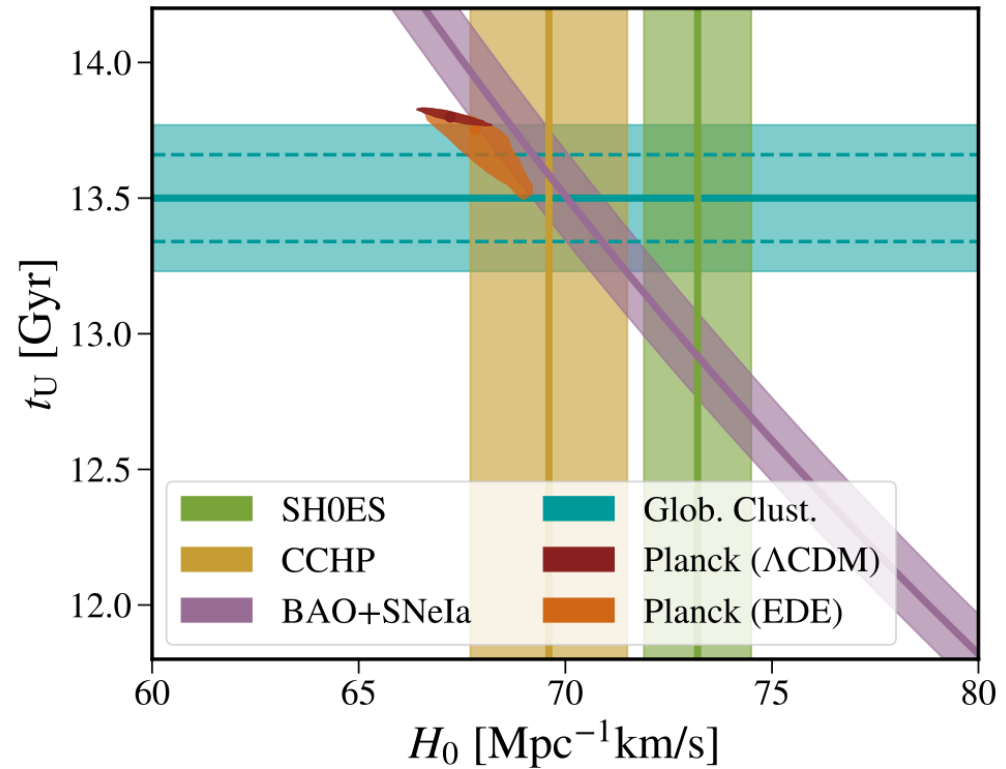
Where is the problem?

$z \sim 0$ measurements do not do assumptions about cosmology

Early-time measurements assume standard Λ CDM.

How can ages help?

Cosmic Complementarity



Bernal et al. PRD 2021

Conclusions

- Model independent determinations of cosmological quantities can be the clue to the **next disruption [be too conservative and you will miss new physics]**
- Stellar ages agree with that derived by Planck18 (LCDM)
- Expansion history is as predicted by LCDM (we showed this 10 years ago)



Elena Tomasetti



David Valcin



Carmela Lardo



Michele Moresco



Ben Wandelt



Fred Courbin



Licia Verde



RJ



Uros Seljak



Andrea Cimatti